B.TECH SEMESTER EXAMINATION - 2023
Department of Computer Science & Engineering
Subject: Operating Systems (CS-401)
Time: 3 Hours Max. Marks: 100

Instructions: All questions carry equal marks.

1. Define Operating System. Discuss the dual-mode operation in modern operating systems. [10 Marks]

2. What is a Process Control Block (PCB)? Explain process states with a state transition diagram. [10 Marks]

3. Explain Deadlock. Discuss the necessary conditions for a deadlock to occur in a system. [10 Marks]

4. What is CPU scheduling? Differentiate between preemptive and non-preemptive scheduling. [10 Marks]

5. Explain the differences between Paging and Segmentation memory management techniques. [15 Marks]

6. Describe Bankers Algorithm for deadlock avoidance with a clear illustrative example. [15 Marks]

7. Explain Page Fault and Page Replacement algorithms: FIFO, LRU, and Optimal. [15 Marks]